

PRAVEEN KUMAR GORREPATI

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EDUCATION

University of Houston

Aug 2025 – 2027

Master's in Engineering Data Science

Houston, TX

- **Relevant Coursework:** Machine Learning, AI Systems, Deep Learning, Probability and Statistics Analysis, Data Mining.

Indian Institute of Information Technology Sri City

Dec 2021 – May 2025

B.Tech in Electronics and Communication Engineering

Sri City, India

- **Relevant Coursework:** Data Structures, Algorithms, Python, Machine Learning, OOP, Computer Networks, Cryptography.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, JavaScript, Bash
- **Machine Learning & AI:** Scikit-learn, XGBoost, PyTorch, TensorFlow, CNNs, YOLOv8, Regression, Classification, Clustering
- **Data Science:** Pandas, NumPy, Data Cleaning, Feature Engineering, Exploratory Data Analysis, Data Quality Assurance
- **Deployment & Tools:** Flask, FastAPI, Docker, Git, Linux, HPC Clusters
- **Visualization:** Power BI, Tableau, Matplotlib, Excel

EXPERIENCE

Asset Management Intern

May 2026 – Aug 2026

Port Houston

Houston, TX

- Structured marine asset inspection follow-up actions into standardized datasets to improve maintenance tracking, backlog visibility, and reporting readiness.
- Reviewed baseline and routine inspection reports to extract asset IDs, components, element types, condition findings, priorities, and recommended maintenance actions.
- Developed a tracking framework to classify open, carried-forward, partially completed, and pending-verification maintenance issues across inspection cycles.
- Explored SQL database tables and supported Power BI and IBM Maximo readiness by aligning inspection findings, maintenance status, asset information, and data quality requirements.

Technical Support Associate

Sep 2025 – May 2026

Hewlett Packard Enterprise Data Science Institute, University of Houston

Houston, TX

- Assisted **500+** students and researchers in developing machine learning workflows using **Python, Scikit-learn, PyTorch**, and **TensorFlow**.
- Provided technical guidance on **data preprocessing**, feature engineering, model evaluation, and research-based ML project development.
- Supported **GPU-based model training** on **Linux HPC clusters** and troubleshoot technical issues related to Python dependencies, ML environments, and HPC resources.

PROJECTS

Deep Learning for Road Infrastructure Monitoring using Computer Vision

- Developed deep learning models using **YOLOv8** to detect potholes and road anomalies from large-scale road imagery.
- Collected and annotated a dataset of **50,000+** road images and **3,000** pothole samples for training computer vision models.
- Evaluated model performance using **precision (93.7%), recall (82.9%), and F1-score (88.2%)** to ensure reliable road defect detection.
- Built a real-time road quality analysis pipeline integrating **multi-sensor data fusion and clustering algorithms** to compute a Road Traffic Quality Index (RTQI).

Machine Learning Analysis of Offshore Oil & Gas Well Performance

- Analyzed **20,000+** wells with 27 features; performed preprocessing, outlier detection (**Z-score**), and feature engineering.
- Built **Random Forest and XGBoost** models to predict drilling efficiency and completion time.
- Identified key operational patterns to support **data-driven optimization** in offshore drilling.

PUBLICATIONS

- **Design of Efficient RAM Cell using Quantum Dot Cellular Automata**, IEEE International Conference on Computer, Electronics, and Electrical Engineering, 2024.

CERTIFICATIONS

- **IoT and Autonomous Systems:** British Council and IEEE ITSS, covering IoT applications and intelligent systems.
- **Full Stack Development:** Completed Udemy course with hands-on experience.
- **Smart Mobility Hackathon:** Innovated road safety technologies including skid detection and pothole detection using IMU sensors; secured leading position.
- **IEEE Conference Participation:** 2nd International Conference on Computer, Electronics and Electrical Engineering, IEEE, 2024.